

Health Impact Functions

Selected Health Impact Functions

Filter Dataset: EPA Standard Health Functions (2021) | Filter Endpoint Group: All | Filter: | ☒ Groups | Add Selected

Endpoint Group	Endpoint	Pollutant	Geographic Area	Author	Start Age	End Age	Beta Distribution	Metric	Season
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Zanobe...	25	44	Normal	D24HourMe...	
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Zanobe...	18	24	Normal	D24HourMe...	
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Zanobe...	65	99	Normal	D24HourMe...	
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Zanobe...	55	64	Normal	D24HourMe...	
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Zanobe...	45	54	Normal	D24HourMe...	
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Zanobe...	25	44	Normal	D24HourMe...	
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Zanobe...	18	24	Normal	D24HourMe...	
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Sullivan...	65	99	Normal	D24HourMe...	
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Sullivan...	55	64	Normal	D24HourMe...	
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5		Zanobe...	65	99	Normal	D24HourMe...	

Selected Health Impact Functions (1)

Endpoint Group	Endpoint	Pollutant	Geographic Area	Author	Start Age	End Age	Race	Ethnicity	Gender	Incidence Dataset
Acute Myocardial Infa...	Acute Myocardial ...	PM2.5	Everywhere	Peters e...	18	24				Other Incidence (201...

Delete Selected | Advanced | Cancel | Save As (*.cfgx) | Run

6.2.3 Advanced Configuration Settings

Clicking the **Advanced** button at the bottom of the **Health Impact Functions** window gives you access to a number of options for tailoring your assessment.

The Point Mode and Monte Carlo options allow you to generate an average incidence estimate, or a range of results that mirror the uncertainty in the inputs to the health impact functions. By default, BenMAP will run in Monte Carlo mode. If you select the option for *Run in Point Mode*, BenMAP-CE uses the mean values of the inputs to the health impact functions, and generates a single "point estimate" of the change in adverse health effects.

With the Monte Carlo (default) option, you can generate a number of estimates that mirror the variability in the inputs to the health impact functions. The Monte Carlo option allows you to generate specific percentiles along the estimated incidence distribution. For example, if you specify **20 Percentiles** (default value), then BenMAP-CE will generate incidence estimates of the 2.5th percentile, 7.5th percentile, and so on, up through the 97.5th percentile. The number of points suggested in the drop down menu for **Percentiles** varies between 10 and 100. The greater the number of chosen

points, the greater the amount of time BenMAP-CE will need to process the results. The relationship between the number of points and time to process is essentially linear, so a doubling of the number of points would double the processing time.

In the Monte Carlo (default) option, the program will perform 10,000 iterations to calculate uncertainty. You may change this number, but the greater the number of iterations, the longer the processing time.

If you choose to *Run in Point Mode*, the field for **Percentile** points for Monte Carlo sampling is disabled and will be ignored (treated as zero). However, with the default Monte Carlo option, the program will still report a point estimate. As discussed in Chapter 7 on Aggregation, Pooling, and Valuation, by choosing the Point Mode, you limit your ability to pool the results. You cannot conduct fixed effect/random effects pooling, or any other procedure that depends on knowing the distribution, or the range of variability of the incidence estimates.

The **Air Quality Threshold** indicates the minimum air quality value that BenMAP-CE will use to quantify health impacts. That is, air quality metrics below the threshold will be replaced with the threshold value. With a threshold of zero, there is no impact on the estimates generated by the health impact functions. However, as the threshold increases, then it will have a progressively larger impact on the incidence estimate. The **Air Quality Threshold** option allows you to explore the impact of any given threshold on the incidence estimate. This can also be useful for scenarios where you might want to know the incidence associated with changes in air quality occurring only above a standard.

If you are estimating impacts for specific population subgroups (e.g., stratification by gender, age, race, or ethnicity), BenMAP-CE provides two options for calculating baseline incidence:

- 1) You can use incidence rates averaged over gender, race, and ethnicity (default). (This option must be selected for the rates supplied with BenMAP-CE, as they are not stratified by gender/race/ethnicity).
- 2) You can use incidence rates that match the gender, race, and ethnicity selected with your health impact function. (You must import stratified rates for use with this option). If you select this option and have not imported rates that match each subgroup, the groups without incidence rates will not be included in the calculations and will have point estimates of zero.

It is recommended that users select this second option if your incidence/prevalence data include overlapping groups. For example, the asthma exacerbation prevalence rate for ages 5-17 has two race groups, one is “ALL” with value 0.107 and the other is “BLACK” with value 0.177.

If the first option (above) is selected, BenMAP will use the arithmetic mean of both values (0.142) as the prevalence rate in its calculations, regardless of whether users select “ALL” or “BLACK” as the race category—or left the race category blank.

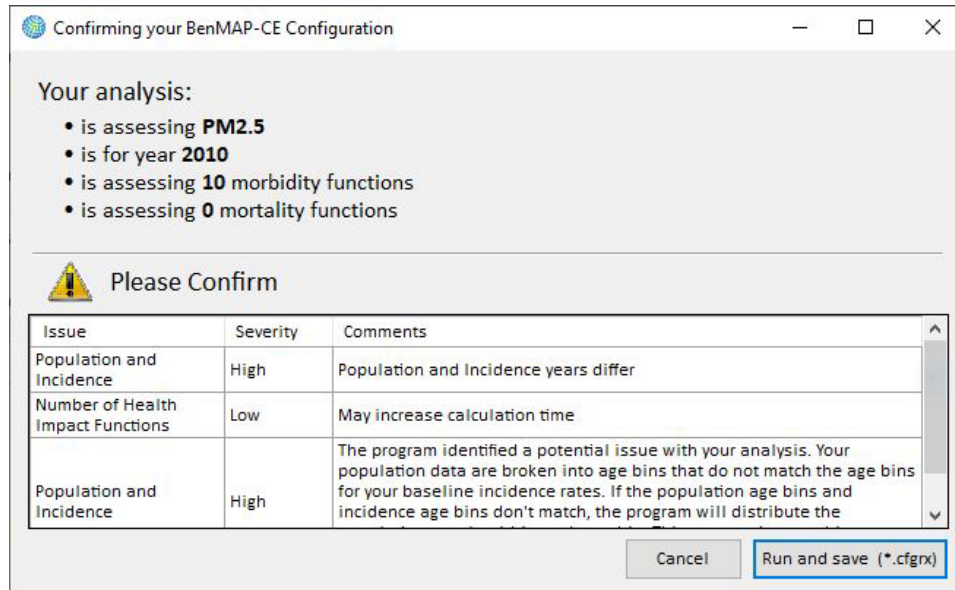
If the second option is selected, and if users select “ALL” as the race category (or leave it blank), BenMAP will first check if this dataset has overlapping race groups to avoid double counting of individuals in incidence or prevalence calculations. In this example, BenMAP will use the “ALL” prevalence rate of 0.107, which exactly matches the name of the selected race. Thus, it is recommended that users select the second option unless the user is certain that incidence or prevalence rate categories do not overlap.

After making selections for calculating impact functions, BenMAP-CE allows you to save the configuration for future use. Click the **Save As (*.cfgx)** button and specify a file with a .cfgx extension. Advanced settings options will be displayed in the Audit Trail.

6.3 Run Health Impact Configuration

To execute the calculation of incidence for the health impact functions in the configuration, click the **Run** button on the bottom right-hand corner of the **Health Impact Functions** window.

A confirmation window will pop up to notify users of key data points and raise potential issues with the run, shown below. To proceed, weigh these limitations and click **Run and save (*.cgrx)**. BenMAP-CE will require that you specify a file in which to save the results, with a .cgrx extension.



6.4 View and Export Health Impact Results

The **Health Impact Results** report gives you the opportunity to examine the incidence results of each health impact function applied at the grid-cell level, or temporarily aggregate them to, say, the state or national level. The configuration results files (*.cfgrx*) contain “raw” health impact estimates that you have not yet aggregated, pooled, or valued.

To begin, click the **Health Impact Results** tab in the upper portion of the main window. To display incidence results for a single study (i.e., health impact function), double-click on the study of interest (or select the study and click the **Show Results** button). Your results will be generated and displayed in a results table (on the **Data** tab). If you choose a single study, you will also have options to view results on the **GIS Map** tab and in a simple bar chart (view in **Chart** tab.) To change the selected study, double-click on a different study choice or select the study and click the **Show Results** button to refresh the display.